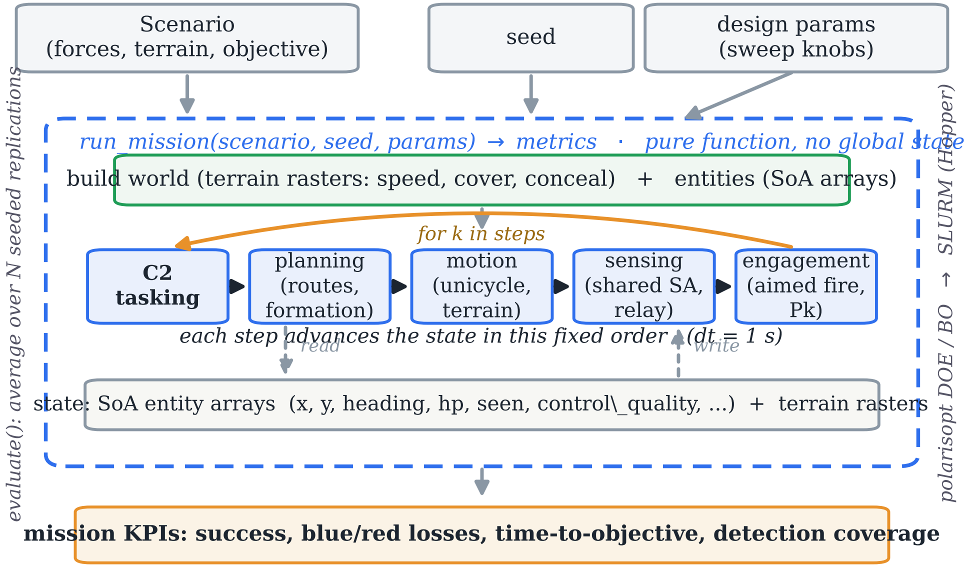
Scenario
(forces, terrain, objective)
seed
design params
(sweep knobs)
run_mission(scenario, seed, params) → metrics · pure function, no global state
build world (terrain rasters: speed, cover, conceal) + entities (SoA arrays)
for k in steps
C2
tasking
planning
(routes,
formation)
motion
(unicycle,
terrain)
sensing
(shared SA,
relay)
engagement
(aimed fire,
Pk)
each step advances the state in this fixed order (dt = 1 s)
read
write
state: SoA entity arrays (x, y, heading, hp, seen, control_quality, ...) + terrain rasters
mission KPIs: success, blue/red losses, time-to-objective, detection coverage
evaluate(): average over N seeded replications
polarisopt DOE / BO → SLURM (Hopper)